`

AL FLASHER 12 V DROP TEST REPORT

Part Name	AL FLASHER 12 V	**Application**	COMMERCIAL APPLICATION
Customer Part No.	FGG00600	**Customer**	ASHOK LEYLAND
UCAL Part No.	RDEPL/AL/15	**Report No**	R&D/AL/12VF/VAL/26/018

INTRODUCTION:

UCAL is currently supplying 12V AL Flashers to Ashok Leyland for commercial applications. This validation plan has been prepared to define the test requirements and validation activities for the current production parts of the respective product. The testing will be carried out as per the approved validation plan to verify product performance, functional reliability, and compliance with specified requirements under defined test conditions.

PURPOSE:

To evaluate the 12 V flashers assembly after **Drop test.**

TEST SPECIFICATION:

The Drop Test shall be carried out under the following test conditions:
The sample shall be tested in non-operating condition without any electrical connection. The unit shall be allowed to fall freely under gravity without applying any external force. One drop on each face of the sample (Total 6 Faces). The unit (AL Flasher 12V) shall be placed in its normal mounting/use position and lifted to a height of 1 Meter ± 10 mm from the test surface. The sample shall then be released to fall freely onto the rigid MS plate surface. After completion of one drop, the unit shall be rotated sequentially so that each adjoining face rests on the test surface, and the dropping procedure shall be repeated until all six faces have been subjected to one free-fall impact.

ACCEPTANCE CRITERIA:

After completion of the Drop Test, the sample shall not exhibit any physical damage such as crack, deformation, loosening of parts, or functional abnormality. The product shall meet all specified functional and performance requirements after testing.

RESULT:

SAMPLE NO	PERFORMANCE	APPEARANCE	JUDGEMENT
S:08	O	O	O

O: OK **X: NG** **Δ: To be confirmed**

CONCLUSION:

The performance of the Flasher assembly is acceptable after test. The samples are met the requirements.

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Prepared By	**Approved By**

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DATE:15.04.2026

PERFROMACE TEST SPECIFICATIONS:

SNO	PARAMETERS	SPECIFICATIONS
1	Operating voltage	8~16 V
2	Test voltage	13.5V
3	Flash rate(Turn signal/Hazard warning	60~120 Flashes/Minute
4	One lamp failure flashing rate	>140 Flashes/Min(or)1.8 times of flash rate whichever is greater
5	Duty Cycle	40% ~60%

TEST SETUP:

TURN SIGNALS & HAZARD SIGNALS:

LEFT SIGNAL	RIGHT SIGNAL	HAZARD SIGNAL
		

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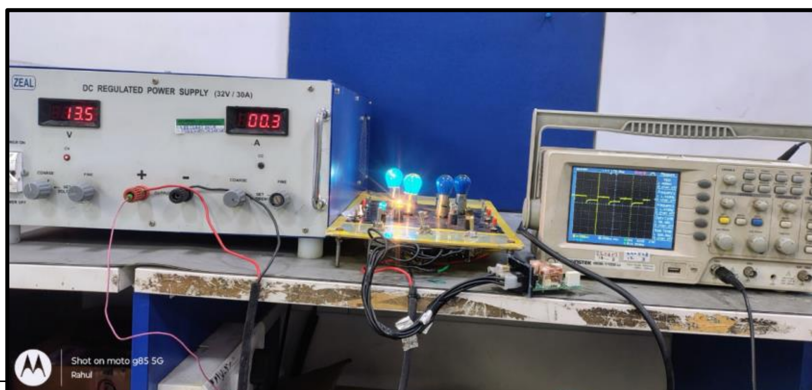
TEST OBSERVATION:

Description	Load	Observations(at 12 V)	Remark
	LED& CLUSTER	Sample No:01	
Turn Signal RH	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	85	OK
Turn Signal LH	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	85	OK
Hazard Mode	2x (2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal)	85	OK
Fault condition Any one 21W Lamp failure	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	170	OK
Duty cycles for RH&LH	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	50%	OK
Duty cycles for Hazard Mode	2x (2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal)	51%	OK

NOTE: The observed flashes are within the specifications & the samples met the performance requirements.

WAVEFORM MEASUREMENT:



TESTED SAMPLES PHOTOS:

BEFORE TEST SAMPLE – S:08	AFTER TEST SAMPLE – S:08
	

OBSERVATION:

There is no significant change in the electrical performance of the flasher samples. The functionality remains satisfactory. The observed flashes are within the specified requirements & the samples met the performance requirements. All function of the Flasher performs as designed before & after the test.